Introduction

A stroke occurs when blood flow to the brain becomes obstructed which results in brain cells receiving insufficient oxygen. This cause permanent brain damage together with long term disabilities. Statistics show that the condition ranks as one of the main deaths causes and disability sources throughout the world. The process of early detection helps to decrease brain damage, which results in better patient outcomes for both survival and recovery.

Medical professionals use neuroimaging methods such as Computed Tomography (CT) and Magnetic Resonance Imaging (MRI) to diagnose brain stroke. Emergency departments use CT scans because they provide quick images which can be accessed easily, while MRI offers superior brain tissue imaging. The emergency department requires immediate patient assessment because a single moment of delay or incorrect reading will result in harm for the patient.

The progress of artificial intelligence has enabled deep learning methods to become effective tools for medical imaging analysis. The Convolutional Neural Network (CNN) architecture enables machines to learn complex image patterns through automatic processes which operate with great accuracy. The models perform automatic detection of hidden defects which require human expertise to identify because they show only minor signs of their presence. The use of deep learning diagnostic systems is growing in clinical settings because they function as medical decision support systems.

The thesis aims to compare different deep learning models which will automatically detect brain strokes through the analysis of neuroimaging data. The study implements VGG16, DenseNet-201, ResNet, and EfficientNet while conducting evaluations of these models through identical testing processes. By comparing their performance, the aim is to identify the most suitable

deep learning model for accurate and efficient stroke detection. The results will help create trustworthy automated diagnostic systems which assist medical professionals while enhancing patient care results.

1.1 Background of the Study

Deep learning techniques have brought about fundamental changes to medical image analysis methods. The traditional machine learning methods depended on expert created features which included texture and shape and intensity values to identify medical image patterns but they struggled to detect intricate patterns. The development of CNNs as deep learning models enables systems to create their own hierarchical feature systems from basic image input.

The diagnosis of brain strokes requires early-stage lesion identification because these lesions exist as small indicators which doctors find challenging to differentiate from standard brain structure. The process of manual diagnosis becomes more difficult because different factors such as image quality problems and noisy elements and variations in patient body structure which exist throughout medical imaging process. The development of automated systems has become essential because they provide doctors with quick and reliable methods to analyze neuroimaging data. The deep learning-based models show high accuracy for identifying CT and MRI scan abnormalities which makes them appropriate for use in stroke detection activities.

Researchers have introduced numerous CNN architectures throughout history which display distinct design elements. VGG16 serves as a widely used baseline model because its architecture combines straightforward design elements with deep structure and identical convolutional layer components. ResNet developed residual connections because it solved vanishing gradient issues, which allowed successful training of extremely deep networks. The

DenseNet-201 architecture enables better information flow between layers because it creates a direct pathway which links all network components together. EfficientNet achieved better model performance because it used balanced approaches to increase network depth and width and resolution.

The models demonstrate successful performance, yet their effectiveness shows considerable differences across different dataset and task combinations. The current research studies use only one architectural design which makes it difficult to evaluate which design performs best. The research needs to compare various deep learning systems through identical dataset testing and standard evaluation methods. This study aims to address this gap by conducting a comprehensive comparative evaluation of popular CNN architectures for brain stroke detection using neuroimaging data.

1.2 Problem Statement

The medical field suffers from persistent problems which make it difficult to detect brain stroke cases accurately and on time. The medical field uses neuroimaging techniques which include CT and MRI scans but their diagnostic value depends on the skills of radiologists who interpret those images. Emergency situations require quick decision making yet manual analysis methods take too much time and make mistakes because of human factors. Medical professionals need to treat patients who show early stroke symptoms because their condition requires treatment within a specific time frame to reduce permanent brain damage risk.

Deep learning methods show good results for medical image classification yet researchers cannot agree on which CNN design achieves optimal performance for brain stroke detection. Existing studies use one deep learning model to evaluate their results which creates challenges for

identifying the most effective and dependable method. Researchers face challenges when comparing study results because of variations in dataset size and preprocessing techniques and evaluation criteria.

The process of choosing a model needs to achieve two requirements which include attaining high accuracy and keeping computational efficiency. Models in actual clinical settings need to deliver their results within short timeframes while operating with minimal computing power. Healthcare facilities find it hard to use complex models which achieve high accuracy because they require extensive resources.

Research problem of this study exists because researchers have not yet created a complete assessment which compares all major deep learning models used for brain stroke detection through neuroimaging data. The study intends to conduct systematic assessments of various CNN architectures using identical testing conditions to determine which model achieves the best performance in accuracy and robustness and operational efficiency.

1.3 Research Questions

I. How effectively can deep learning models detect brain stroke from neuroimaging data?

II. How do VGG16, DenseNet-201, ResNet, and EfficientNet compare in terms of classification performance?

III. Which model offers the best balance between detection accuracy and computational efficiency?

IV. What are the strengths and limitations of each selected architecture for brain stroke detection?

1.4 Research Objectives

The main goal of this research project is to create an automated system which uses deep learning technology to identify brain strokes through analysis of neuroimaging data. The research will test various convolutional neural network models which include VGG16, DenseNet-201, ResNet, and EfficientNet through a unified experimental framework. The research team will assess these models through standard performance metrics which include accuracy and precision and recall and F1-score to determine how well each model performs. The second research objective requires scientists to assess model performance by testing different models' ability to function in real clinical settings. The research develops a deep learning model which detects brain strokes with high accuracy and efficiency while providing research insights that support future research and clinical decision support system development.

Literature Review

2.1 Overview of Stroke

A stroke is a critical nervous system issue that happens when there is a sudden stop or alteration of blood circulation to the brain, which results in an abrupt lack of oxygen and nutrients in that particular part of the brain. The brain cells that have been cut off from the blood supply will die very quickly through the process of necrosis, and if the medical intervention is not fast enough the individual may suffer either permanent nerve injury or die. Moreover, stroke has been one of the top reasons for death and disabilities needing long-term care globally, which makes it a serious problem for the health care system.

Strokes can be divided into two principal types: ischemic and hemorrhagic. Among them, ischemic stroke is the one that constitutes a greater part about 80-85% of the whole stroke case picture. This type of stroke happens when a blood vessel that feeds the brain gets occluded, which is usually brought about by a blood clot or accumulation of fatty substances in the vessel. The result is that the blood flow is cut off and the brain tissue gets damaged. Ischemic strokes can be classified into two more categories: thrombotic and embolic, based on the location of the blockage. A hemorrhagic stroke happens when a blood vessel in the brain ruptures, causing blood to leak into or around the brain thus. This type of stroke is less common but always more severe and has a higher death rate. Depending on the location of the bleeding, generally, hemorrhagic strokes are classified as intracerebral hemorrhage or subarachnoid hemorrhage. The major factors that lead to the occurrence of hemorrhagic strokes are high blood pressure, aneurysms, and head traumas.

Besides these two major forms, there is a transient ischemic attack (TIA) which is sometimes called a "mini-stroke", a temporary condition that occurs when there is a short

interruption of blood flow to the brain. TIAs, although the symptoms go away very quickly, are considered to be an important sign of future strokes and hence, need immediate medical intervention. It is very important to know the various types of strokes to achieve correct diagnosis and treatment planning. Imaging characteristics and treatment options are different for each stroke type, therefore, automated systems for detection have to be developed in a way that they would be able to tell these differences easily, especially in the very beginning stages.

2.2 Pathophysiology of Brain Stroke

The sequence of biological and cellular events that comprise the pathophysiology of a brain stroke occurs during the abnormality in blood flow to the brain. The reliable working of the brain relies on a constant stream of oxygen and glucose. When this stream is interrupted, even for just a few moments, the brain cells begin to lose their function and may end up being irreversibly damaged.

In an ischemic stroke, a blood vessel blockage suddenly reduces cerebral blood flow, bringing about the oxygen and glucose deprivation that miserably dying cells in the brain energies. This initiates the process of cell death, loss of ion balance, and accumulation of toxic substances. Neurons in the impacted area get electrically inactive and may die if blood flow restoration is not done instantly. The central part where there is serious damage is called the infarct core, while the outer part, known as the penumbra, consists of tissue that has been partially damaged but is still capable of being saved.

Hemorrhagic stroke has an entirely different pathological process as compared to ischemic stroke. When a blood vessel ruptures, blood flows into the brain tissue, therefore causing a build-up of pressure inside the skull. This pressure can push the neighboring brain areas and also disturb

the regular blood flow. Moreover, the extravasated blood activates the inflammatory responses and causes more cell death. The combined effects of these factors can very quickly escalate the destruction of the nervous system and the likelihood of complications.

In both kinds of stroke, there are primary injury mechanisms such as inflammation, oxidative stress, and cell death pathways, which are quite complex, and they are termed secondary injury mechanisms. These processes are not confined to the initial event alone but continue and contribute to brain damage progression in a time-dependent manner. The injury is not uniform and its size is determined by a combination of factors such as the area of the stroke, how long the blood flow was cut off, and the patient's general health condition. A good grasp of the pathophysiology of brain stroke goes hand in hand with detection and treatment being done at an early stage. The minor physiological changes that take place during the very initial period of stroke can be seen using neuroimaging techniques. Proper identification of these changes is crucial for the intervention to be timely and it also forms the foundation for the creation of automated systems for stroke detection.

2.3 Neuroimaging Modalities for Stroke Diagnosis

Neuroimaging has a significant role in the diagnosis and treatment of brain stroke. The use of imaging methods enables the doctors to see the brain structures, to point out the areas with problems, and to know the stroke's type, location, and intensity. Fast and precise imaging is very important for the selection of the right treatment methods and for the reduction of brain damage. From the different imaging techniques, Magnetic Resonance Imaging (MRI) and Computed Tomography (CT) are the ones that are most often used in clinical practice.

The above-mentioned imaging methods generate information that supports each other. In the emergency environment, CT is the most used method because of its quickness and wide availability, while MRI has the advantage of being the most sensitive for spotting the very first and mildest ischemic changes. Usually, automated stroke detection systems will use these imaging modalities to extract relevant features for the purpose of classification and analysis.

2.3.1 Magnetic Resonance Imaging (MRI)

Magnetic Resonance Imaging is a non-invasive imaging method that produces detailed images of brain tissue using strong magnetic fields and radio waves. MRI is the technique of choice in early ischemic stroke detection owing to its ability to spot very slight changes in the brain structure and composition of the tissue. Advanced MRI sequences are capable of indicating spots with blood flow restriction and cellular death that other imaging modalities may never show.

MRI has a high spatial resolution and also provides great contrast between soft tissues which makes it a good candidate for the detection of small lesions and separating healthy and affected brain areas. But generally, MRI scans are of a longer duration and the availability of MRI machines in emergency departments is not always the case. On the other hand, the intricacy and price of MRI machines also prevent their use in some areas of medical care.

Nevertheless, these drawbacks, MRI continues to be an essential instrument for stroke diagnosis and is heavily used in clinical trials for automation detection methods' development and evaluation.

2.3.2 Computed Tomography (CT)

Computed tomography (CT scan) is one of the most preferred techniques for diagnosing stroke cases, particularly in emergencies. A CT scan produces a series of X-ray images or slices of the brain and is very effective in quick identification of hemorrhagic strokes. The combination of speed and universal availability of CT has secured its position as the main first-line imaging modality in the assessment of acute strokes.

CT scans are the deciding factor in determining whether there is bleeding in the brain or not before going ahead with the treatment of ischemic stroke. On the other hand, the signs of early ischemia are sometimes not very clearly seen on standard CT images; hence, the diagnosis might be difficult. Picture quality in some cases may also suffer due to noise and low contrast.

Yet, CT still is and will continue to be the primary radiation technique for stroke detection. Its quick imaging time and the fact that it is constantly available in the clinic make it a perfect candidate for deep learning model-based automated analysis, especially in critical situations where time is of the essence.

2.4 Image Processing Techniques in Medical Diagnosis

The application of image processing techniques in medical diagnosis is extensive. Their main purposes are to improve the quality of the images, to extract important information, and to assist in making correct clinical decisions. The medical imaging process usually includes raw images that are not ready for diagnosis because they contain noise, have low contrast, or there is irrelevant background information present. By image processing, the accuracy of the diagnosis is enhanced as the important anatomical structures and pathological regions are made more visible. Some of the preprocessing techniques that are commonly used are noise reduction, contrast

enhancement, normalization, and image resizing. Noise reduction methods are applied to get rid of the unwanted distortions that could have been introduced during the process of acquiring the images. On the other hand, one of the benefits of contrast enhancement technology is that it allows even the dimmest features in the image to come out gradually. Normalization is a technique used to equalize the brightness levels of different images, and this is especially important in the case of data coming from different imaging devices or settings.

Segmentation is a significant image processing technique that is applied in medical diagnosis. This technique, which separates regions of interest, organs or abnormal tissues, from the background, is widely used in stroke analysis. The isolation of segmentation could lead to an easier studying of lesion size, shape, and location. Manual assessment and automated analysis both depend on accurate segmentation.

Feature extraction techniques are applied to recognize the most relevant patterns and characteristics from the medical images. The traditional methods rely on handcrafted features like texture, shape, and intensity-based measures. These methods, however, need domain expertise and may not be very good at generalizing across different datasets. Data augmentation techniques are also among the methods used in image processing, mainly to increase the machine learning model training data diversity. Augmentation methods like flipping, rotation and scaling help build model robustness and lower overfitting.

Taking everything into account, image processing techniques are no less than a major determinant in the medical diagnosis sector as they not only improve image quality but also allow for efficient analysis. They are the cornerstone of deep learning-based systems, where the high-quality input data greatly boosts the model performance and trustworthiness.

2.5 Deep Learning in Medical Imaging

The usage of deep learning in medical imaging has significantly increased due to its capability to automatically discover intricate patterns in huge sets of data. Deep learning models, in contrast to the traditional machine learning methods that rely on human ingenuity for feature design, can handle the medical images, for example, CT and MRI scans, in a straightforward manner. Thus, they are able to record even the most delicate differences in intensity, texture, and structure, which might not be noticed by the human eye.

In the field of medical diagnostics, deep learning methods are employed for tasks like image classification, segmentation, and disease detection. For example, they are very useful in the case of neurological imaging where brain structures are of high complexity and the related changes are often very little in the early stages of the disease. Deep learning systems are capable of processing a large amount of imaging data very fast and with high consistency, thus being able to take part in applications where time is crucial, like diagnosis of stroke.

A further benefit of deep learning technology is that it can facilitate a direct and complete mapping of the images to the diagnostic outputs, thus the end-to-end learning. Transfer learning on the other hand, increases the accuracy of the results by training the medical models on smaller datasets that come from the already pre-trained model on a large generic dataset. In spite of the challenges related to computational cost and interpretability, deep learning is still one of the main contributors to the development of automated medical diagnosis.

2.6 Convolutional Neural Networks (CNNs)

Convolutional Neural Networks (CNNs) can be considered the deep learning models that have a specific application in image processing. The CNN architecture possesses many layers, including convolutional, pooling, and fully connected ones. All layers are built with the same purpose of extracting the spatial characteristics of the image automatically. The layers take their different roles in the entire process of feature extraction, and the end result is the building of a feature hierarchy where the basic edges are first recognized and subsequently combined into more complex features such as different parts of a body.

In the field of medical imaging, the usage of CNNs is particularly prevalent due to the fact that they keep the spatial relationships in the images intact, which is very important in the case of brain scans. CNNs minimize the need for manual feature extraction because they are able to learn the most discriminative features straight from the data. Therefore, CNNs are very useful in the processes of disease classification and lesion detection.

Furthermore, CNNs are very resistant to changes in image size, noise, and intensity, even more so when they are used with preprocessing and data augmentation strategies. Their power of generalization over a variety of imaging conditions makes them appropriate for real-world clinical applications. Consequently, CNN-based models constitute the majority of deep learning systems that are developed for automatic stroke detection using neuroimaging data.

2.7 Transfer Learning in Medical Image Analysis

Transfer learning has become a widely adopted technique in medical image analysis, particularly in deep learning applications where labeled datasets are limited. Transfer learning uses a pre-trained model which has been developed through training on a large dataset that includes

ImageNet to train deep neural networks for specific medical tasks. The model can use its previous knowledge about basic image features which include edges and textures and shapes to analyze medical images at their fundamental level.

The work of collecting and labeling large medical datasets for brain stroke detection research faces two main challenges which include privacy concerns and the need for costly specialized labeling. The implementation of transfer learning enables systems to reach effective performance levels with minimal requirements for data input. Pretrained models such as VGG16, ResNet, DenseNet-201 and EfficientNet have shown strong generalization abilities after they were fine-tuned on medical imaging datasets.

Transfer learning uses two main methods which are feature extraction and fine-tuning. Feature extraction uses a pretrained model that functions as an unchangeable feature extractor which enables training only the final classification layers on new data. Fine-tuning allows model adaptation to the target task through retraining some or all pretrained layers together with the classifier. Medical imaging specialists prefer fine-tuning because it enables model training of specific task features while maintaining advantages from pretrained expertise.

The process of transfer learning allows medical facilities with limited resources to achieve better results because it reduces both their training time and their need for computational power. The process of selecting appropriate pretrained models together with tuning methods needs to receive proper execution because it determines the occurrence of either overfitting or negative transfer. Researchers use transfer learning to build deep learning systems which achieve high efficiency and accuracy in brain stroke detection from neuroimaging data.

2.8 Review of Deep Learning Models Used for Stroke Detection

Numerous studies have explored the application of deep learning techniques for automated brain stroke detection using neuroimaging data. The initial research examined standard CNN models which processed CT and MRI data to identify stroke patients and non-stroke patients. CNN-based methods demonstrate better accuracy than traditional machine learning methods because they automatically extract vital features from raw image data.

Researchers now investigate modern architectures which include ResNet, DenseNet and EfficientNet. The usage of ResNet-based models has become common because they enable training of deeper networks which maintains model performance while enhancing detection precision. DenseNet architectures achieve successful small stroke lesion identification because they enable detailed feature extraction through their design which permits information reuse between different network layers. EfficientNet has gained attention for achieving high accuracy with fewer parameters which makes it suitable for real-time applications.

The majority of current research studies show positive results because they only assess one deep learning model which makes it impossible to compare how different architectures perform. Researchers face difficulties when trying to compare their studies because the studies use different datasets and preprocessing methods and evaluation criteria. The research achieves high accuracy results, but it fails to present a thorough assessment of sensitivity together with false-negative rates which hold vital importance for clinical stroke diagnosis.

The current limitations need researchers to conduct a complete evaluation which tests multiple deep learning models under the same experimental conditions. The approach enables

researchers to evaluate model performance more objectively while they gain understanding about which architectural designs work best for stroke detection.

2.9 Challenges in Automated Brain Stroke Detection

The process of establishing automated brain stroke detection through deep learning technology encounters multiple obstacles which decrease both its operational effectiveness and its medical use. The system encounters its primary obstacle because there exist insufficient labeled neuroimaging datasets for research purposes. The process of medical image annotation requires radiologist professionals, who create high costs and extended time periods for acquiring medical imaging data.

The problem of class imbalance constitutes an important challenge because stroke cases appear less frequently than normal imaging samples. The existing situation leads models to display a tendency toward predicting non-stroke outcomes which results in decreased detection rates. The various clinical settings result in problems for model accuracy because they use different imaging methods and scanning devices which produce images of varying quality.

The detection of minor stroke indicators becomes challenging because the subtle features of CT and MRI scans get concealed by noise and low contrast and various artifacts. The deep learning models require extensive processing power which does not exist in all healthcare facilities.

Deep learning models face an additional problem because their operation remains hidden from view. Clinicians need to understand how the system reaches its predictions before they will trust the automated system. Reliable stroke detection systems need to solve these problems before they can be used in real-world situations.

2.10 Summary of Past Works and Accuracy

References	Models	Dataset	Results
Yelken & Ceylan (2025)	ResNet-50, ResNet-101, ResNet-152	Brain CT scans (dataset from Turkish Ministry of Health)	ResNet-152 (97.24%)
Tursynova & Omarov (2024)	(CNN + EfficientNetB7 + DenseNet201)	Kaggle stroke dataset	94.67%
Korde et al. (2024)	(HLTDS) vs. VGG16, ResNet50, DenseNet, EfficientNet	Brain CT images	96.45%
Ozaltin et al. (2022)	OzNet (Novel CNN) + mRMR feature selection	Brain CT images	98.42%
Sailasya & Kumari (2021)	Traditional ML	Dataset of physiological factors	82%

Methodology

The research presents an automated system which utilizes deep learning methods to identify brain strokes through analysis of neuroimaging data. The methodology enables researchers to evaluate different convolutional neural network designs because it maintains identical testing conditions which guarantee unbiased results and study reproduction. The complete system includes six components which are data acquisition and preprocessing and model implementation and training and evaluation and comparative analysis.

Clinical annotations serve as the basis for labeling images into two categories which are stroke and non-stroke. Image preprocessing is done before model training to enhance image quality and create standardized input dimensions which all models will use. The process functions as an essential procedure which decreases background interference while it enhances visual distinction and establishes standard intensity levels for images.

The workflow uses four deep learning models VGG16, DenseNet-201, ResNet and EfficientNet. The networks improve their performance on medical datasets through transfer learning which uses pretrained weights to start model training. The training strategy remains identical for all models because the researchers use the same data splits and batch sizes and optimization methods.

The evaluation of model performance uses standard classification metrics which include accuracy and precision and recall and F1-score and confusion matrix analysis. The research analyzes two aspects of computational efficiency which include training time and inference performance evaluation. The results are then compared to identify the most effective model for

brain stroke detection. The experimental outcomes show robustness through the design which maintains fairness and clinical relevance.

3.1 Workflow Diagram

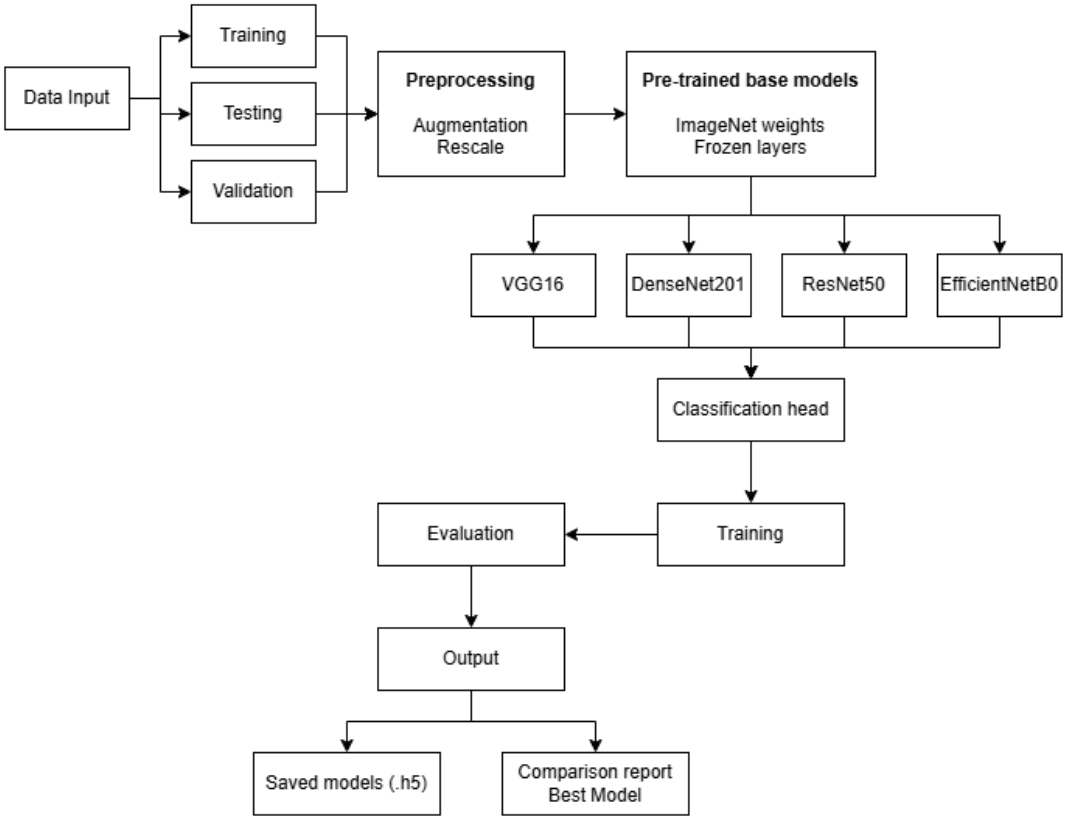


Fig: Proposed Framework

3.2 Dataset

The dataset used in this study consists of labeled neuroimaging scans collected directly from collaborating hospitals, ensuring clinical relevance and real-world applicability. The dataset includes brain CT images representing both stroke-affected and normal cases. CT imaging is

selected as it is the primary diagnostic modality in emergency clinical settings, particularly effective for the rapid detection of hemorrhagic stroke.

The dataset comprises scans from multiple patients with varying brain anatomies, imaging conditions, and stroke characteristics. This diversity introduces a wide range of imaging patterns, which is essential for improving the generalization capability of deep learning models.

The dataset is divided into three parts which include training, validation and testing to maintain consistent experimental conditions. The training set is used to learn model parameters, and the test set is reserved for final performance evaluation. The experiments use a constant data split ratio which enables model comparisons to be conducted in an impartial manner.

Medical imaging tasks often face difficulties because their datasets contain unbalanced class distribution. Data augmentation method applied to produce additional training samples which reduced model bias through increased sample diversity. The researchers developed the dataset selection and preparation methodology to support reliable evaluation of deep learning-based stroke detection systems.

3.3 Data Preprocessing

The process of data preprocessing serves as an essential procedure for transforming neuroimaging data into a format suitable for deep learning model training. The unprocessed CT images contain various disruptions and changes in image brightness and unneeded background elements which together create obstacles that hinder model effectiveness. The research team uses multiple preprocessing methods which they apply to enhance image quality while making images suitable for CNN system requirements.

The team first resizes all images to a specific resolution that meets the input standards of their chosen pretrained models. Through pixel intensity normalization the process establishes image values at a fixed range which helps to maintain training stability during model development. Noise reduction methods allow the removal of undesired visual elements while the process simultaneously enhances the visibility of major anatomical characteristics.

The training dataset size expands through data augmentation methods which create additional data for training while helping models to perform better on unseen data. The methods use horizontal and vertical image flipping together with rotation and zooming and scaling techniques to create their effects. Through augmentation models achieve better performance because it helps them to learn features which remain unchanged across various image display methods and environmental conditions.

The team uses class balancing techniques to correct problems that arise from unbalanced dataset distribution. Preprocessing ensures that all models receive identical input data which enables researchers to evaluate model performance across different architectures in a standardized way.

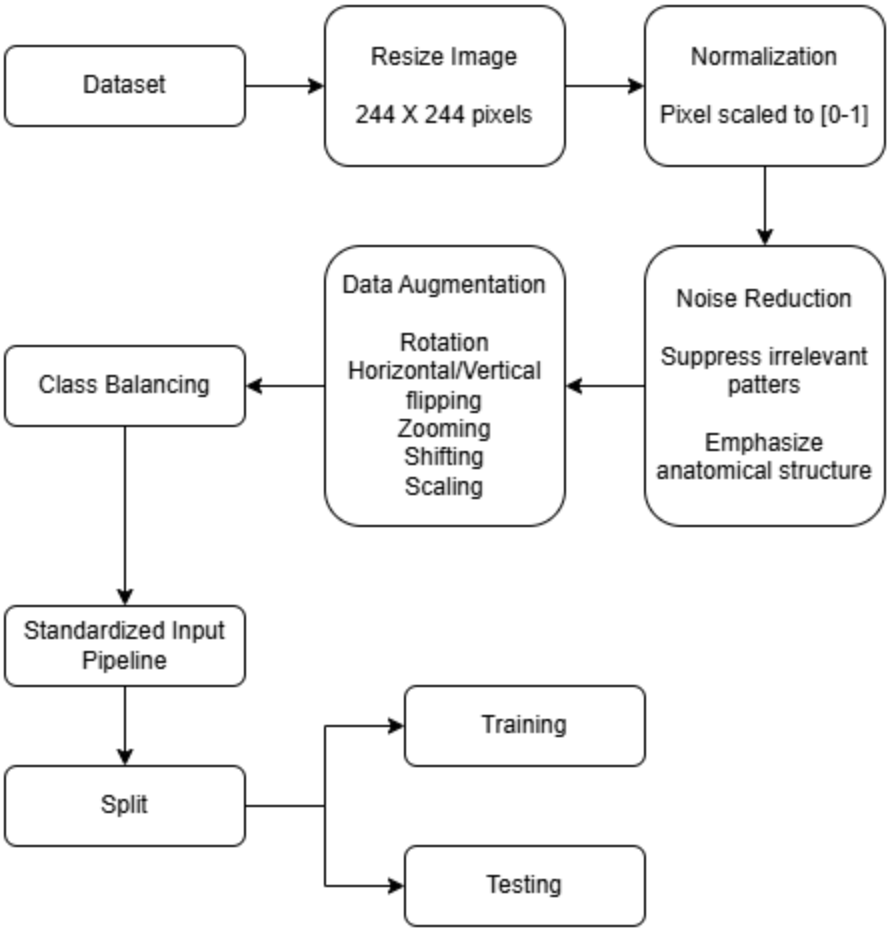


Fig: Data Preprocessing

3.4 Classification Head

The classification head is the final component of a convolutional neural network (CNN) that transforms extracted features into output predictions. It follows the feature extraction backbone and is responsible for classifying medical images into categories such as *stroke vs. non-stroke*.

If ***f*** represents the feature vector from the backbone:

$$y = \sigma (Wf + b)$$

where **W** and **b** are learnable parameters and σ is the activation function.

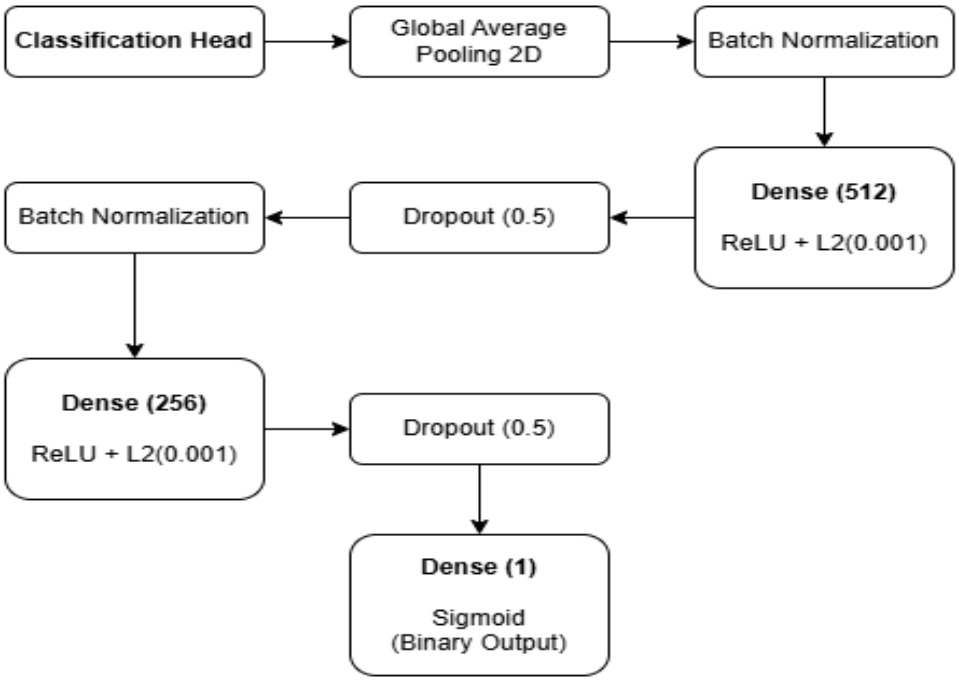


Fig: Classification Head

3.5 Deep Learning Model Architectures

This study evaluates four widely used deep learning architectures for brain stroke detection: VGG16, DenseNet-201, ResNet, and EfficientNet. Each architecture has unique structural characteristics that influence feature extraction and classification performance. All

models were adapted using transfer learning, starting with pre-trained weights and adding custom classification layers for stroke detection.

3.5.1 VGG16

VGG16 uses its basic architecture which contains a series of convolutional layers followed by fully connected layers to create its design. The system achieves its strength through its consistent use of small filters across all architectural components. The system provides strong interpretative power through its basic design which functions as an effective standard model. VGG16 requires extensive memory and processing power because its deep structure with multiple layers and high number of parameters. The system performs well for various tasks but it does not provide optimal performance when resources are limited and fast processing is required.

3.5.2 DenseNet-201

DenseNet-201 establishes a new approach to layer connectivity through its design because all layers in the network connect to each other using feed-forward connections. The dense connectivity system enables complete feature reuse, which empowers the network to retain and utilize data from its previous processing stages. The design solution solves the vanishing gradient issue while improving the system's ability to identify difficult-to-detect advanced features. The system needs large amounts of memory because it stores all feature maps processed between different stages of operation. The system can achieve better accuracy for detailed tasks, but it faces deployment difficulties on devices that have restricted memory resources.

3.5.3 ResNet

The residual block of ResNet serves as its fundamental breakthrough because it uses skip connections to transmit information through its one or more-layer shortcuts. This solution efficiently addresses the vanishing gradient problem, which enables deep networks with hundreds of layers to undergo stable training. The network can learn identity functions through its residual connections, which supports better results during complicated tasks. The system achieves higher efficiency than VGG16 through its depth and residual block design, but it requires complex architectural elements. The system establishes powerful capabilities, but its decision-making processes become difficult to understand because of its intricate internal mechanisms.

3.5.4 EfficientNet

EfficientNet uses a compound scaling method to achieve balanced network depth and width scales together with different input image sizes. The system follows a structured method which operates efficiently to reach its highest performance level within given computing limits. The system delivers exceptional accuracy results while using significantly fewer system resources than competing architectural designs which offer similar performance capabilities. The design excels in applications which demand simultaneous management of operational efficiency and resource consumption and accuracy in forecast results. The system's efficient design structure requires multiple processing layers which decrease the system's straightforwardness when compared to traditional linear system designs.

3.6 Model Training Strategy

The training process achieves consistent results through its implementation of fairness measures for all deep learning models. The model training process uses identical dataset partitions

and preprocessing methods and training specifications. The stroke detection task requirements are met through transfer learning which freezes the initial layers while enabling fine-tuning of higher layers.

The models are trained using a suitable optimization algorithm such as Adam, with a predefined learning rate and batch size. The model optimization process uses binary classification loss functions which are suitable for this purpose. The implementation of early stopping and model checkpointing methods protects against overfitting while maintaining the best model weight configuration. The unified training procedure enables performance evaluations because it maintains consistent training conditions which only change when the system architecture is modified.

3.7 Performance Evaluation and Comparative Analysis

Model performance is evaluated using a set of metrics, including accuracy, precision, recall, F1-score, and confusion matrix analysis. These metrics provide insights into both overall classification performance and class-specific behavior. Recall is given particular importance due to its clinical relevance in detecting stroke cases. False-negative predictions are carefully analyzed to assess potential diagnostic risks. Computational efficiency metrics such as training time and inference speed are also considered. A comparative analysis is conducted to assess the strengths and limitations of each model. This evaluation framework enables the identification of the most effective deep learning architecture for automated brain stroke detection.

3.8 Implementation Environment

The proposed framework uses Python programming together with deep learning libraries from TensorFlow and PyTorch for its implementation. The system with GPU support conducts

experiments to speed up the process of training machine learning models. The research uses standard data processing libraries together with visualization libraries and evaluation libraries to achieve reproducible results. The researchers conducted all experiments using the same hardware and software setup to ensure consistent evaluation of all models.

3.9 Evaluation Metrics

Evaluation metrics are essential for measuring the performance of the proposed model. Multiple metrics were applied to ensure a comprehensive and unbiased evaluation of each architecture. These metrics support comparative analysis by highlighting differences in model behavior, reliability, and overall effectiveness. By using a standardized evaluation approach, the study ensures that the results are fair, reproducible, and suitable for identifying the most appropriate model for practical deployment.

3.9.1 Accuracy

Accuracy can be described as the proportion of the total correct predictions which were predicted by the model. Although it provides a general evaluation of model performance, it is deceptive whenever the data is imbalanced.

$$\text{Accuracy}=\frac{TP+TN}{TP+TN+FP+FN}$$

TP	True	Positives
TN	True	Negatives
FP	False	Positives

FN	False	Negatives

3.9.2 Precision

Precision determines what percentage of predicted positive observation are correct, among the total number of predicted positive observations. It is especially essential in health care, in which it is very important to avoid false positive results (diagnosis of a healthy person who has a tumor).

$$\text{Precision}=\frac{TP}{TP+FP}$$

3.9.3 Recall

Recall refers to the possibility of the model to recognize all adequate cases of a condition. In this regard, it determines the proportion of the actual number of tumors that were appropriately identified and thereby hinders false negatives.

$$\text{Recall}=\frac{TP}{TP+FN}$$

3.9.4 F1 Score

The harmonic mean of precision and recall is F1 score. It can reasonably handle the trade-off between false positives and false negatives and this makes it particularly useful in medical imaging work where both forms of mistakes are important.

$$\text{F1-Score}=2\times\frac{Precision\times Recall}{Precision+Recall}$$

3.9.5 Confusion Matrix

A confusion matrix provides an inclusive visualization of the classificatory performance of the model showing the number of correct and false findings of each class. It will be possible to identify the pattern in misclassifications and enable one to assess what types of tumors are more likely to be confused with others.

3.10 LIME Explanation

LIME (Local Interpretable Model-agnostic Explanations) is used to explain the predictions of complex deep learning networks. It focuses on a single input instance and analyzing how small changes in the input affect the model's output. For image data, LIME perturbs different regions of the image and identifies which areas have the most influence on the prediction and highlights these important regions, helping users understand why the model made a specific decision. Due to LIME's model-agnostic behavior it can be applied to any classifier. In medical applications such as stroke detection, LIME is especially valuable because it improves transparency and helps clinicians trust the model's decisions by showing the relevant regions in the scan.

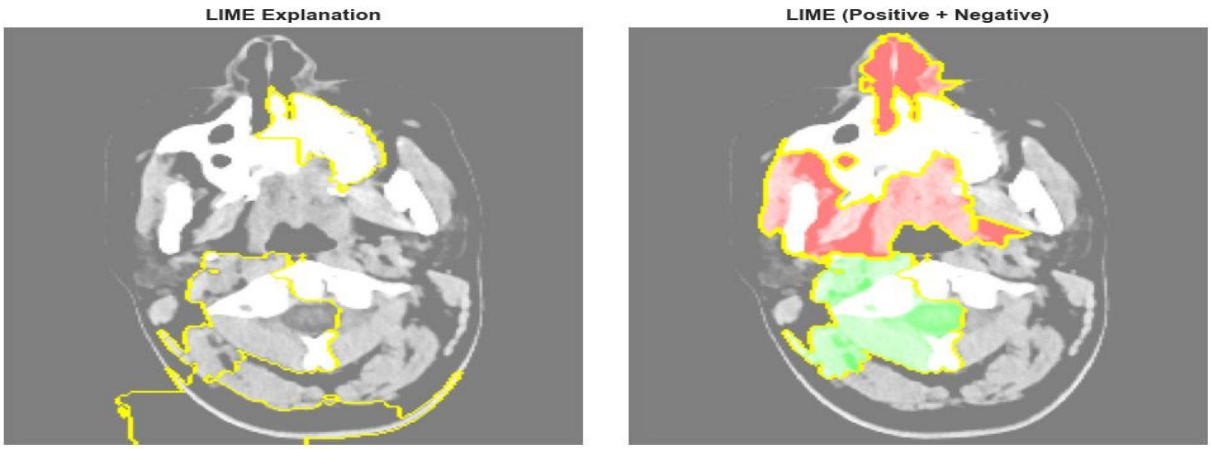


Fig: LIME Explanation during Model training

Highlighted regions in the given left image shows key areas used by the model while the right-side highlights two different component. The red regions show the Positive contribution (support prediction) and the Green regions shows the Negative contribution (oppose prediction)

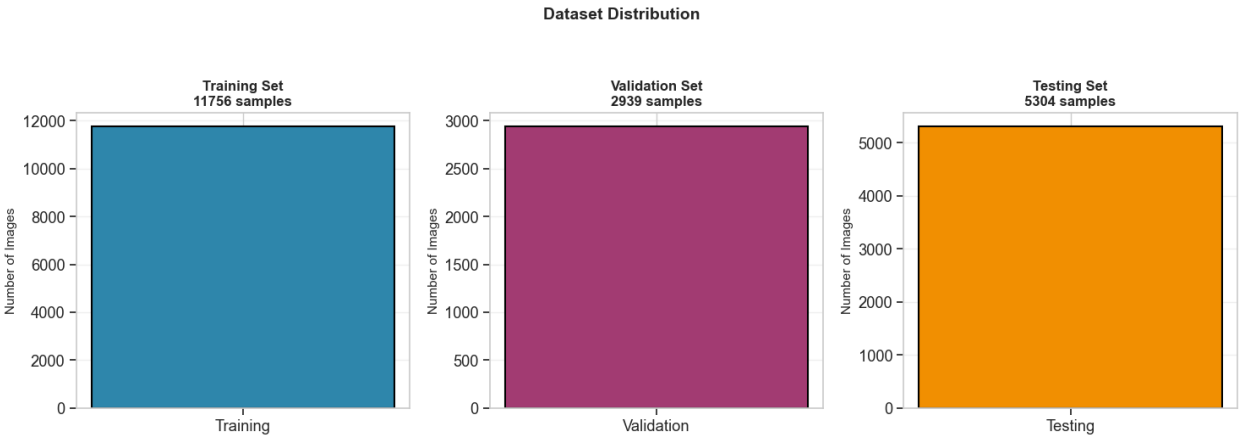
Result and Discussion

This chapter presents the experimental results obtained from the implementation of deep learning models for automated brain stroke detection using CT neuroimaging data. The models were trained and evaluated using the dataset described in Chapter 3. The goal of the experiments was to measure how effectively the proposed system can classify images into normal and stroke categories.

The workflow included data loading, preprocessing, augmentation, model training, validation, and final performance evaluation on unseen test data. Several metrics such as accuracy, precision, recall, F1-score, and ROC curves were used to assess model performance.

4.1 Dataset Distribution

Before training, the dataset was divided into training, validation and testing subsets. The training set was used for learning model parameters and the test set evaluated generalization capability.



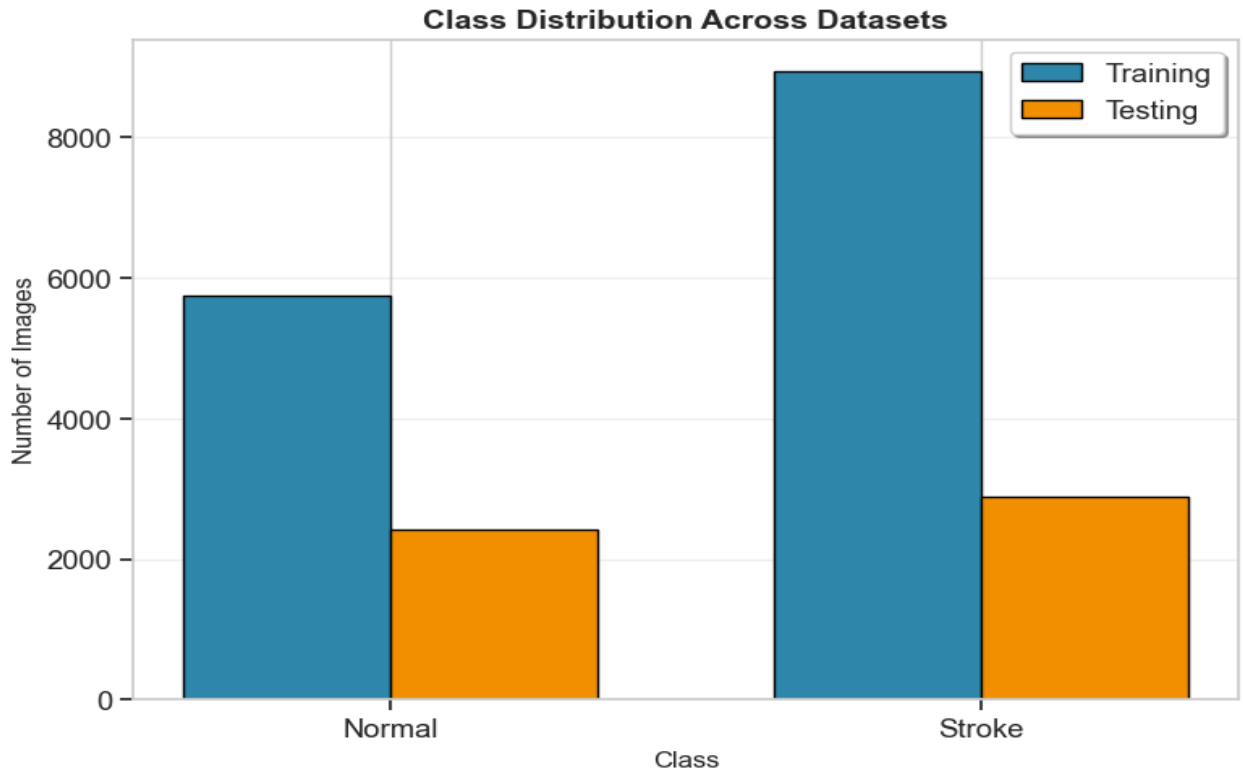
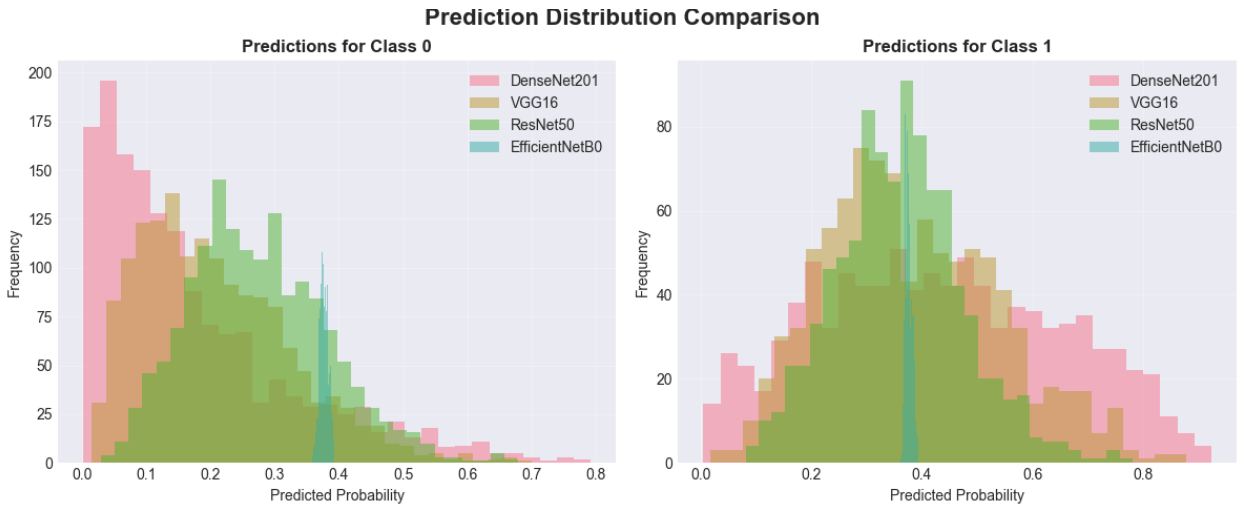


Fig: Class Distribution across Dataset

4.2 Prediction Distribution Comparison

Prediction distribution helps analyze model bias and class balance across predictions. It helps visualize the balance between predicted classes across different models. Differences in distribution indicate potential bias or overfitting toward a specific class. A well-balanced distribution suggests better generalization and reliable performance.



4.3 Confusion Matrix

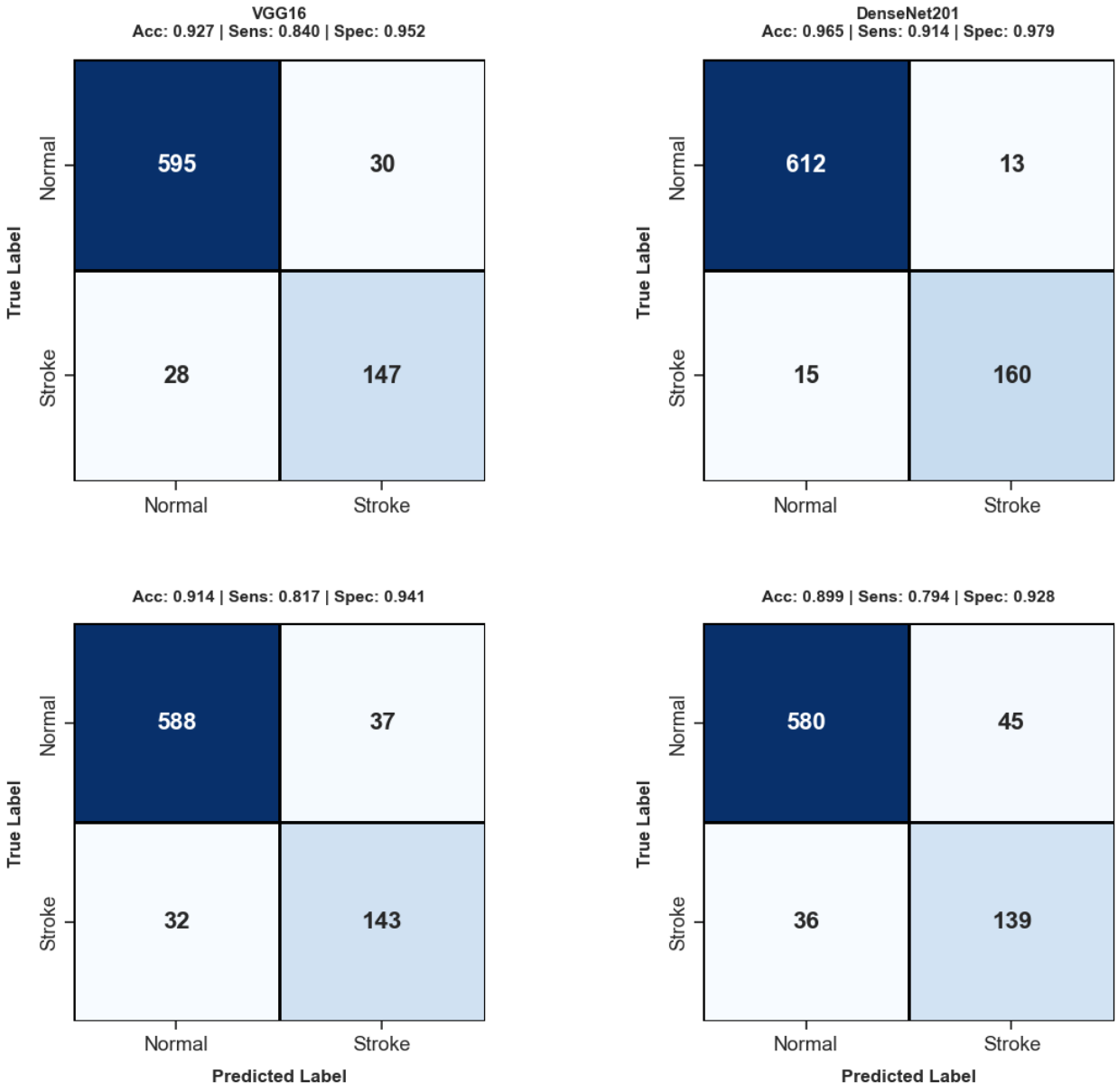
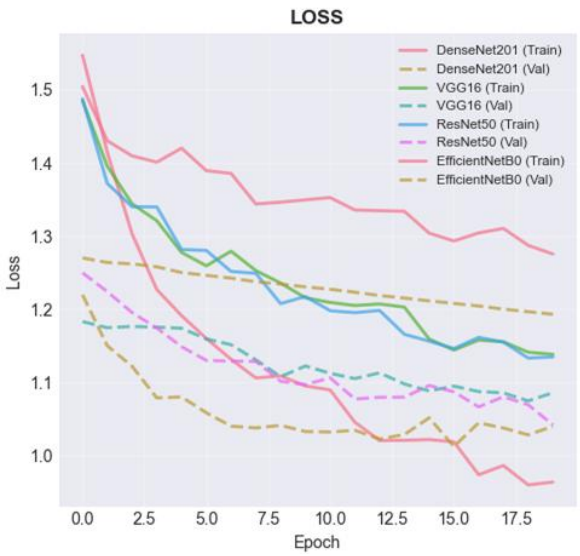


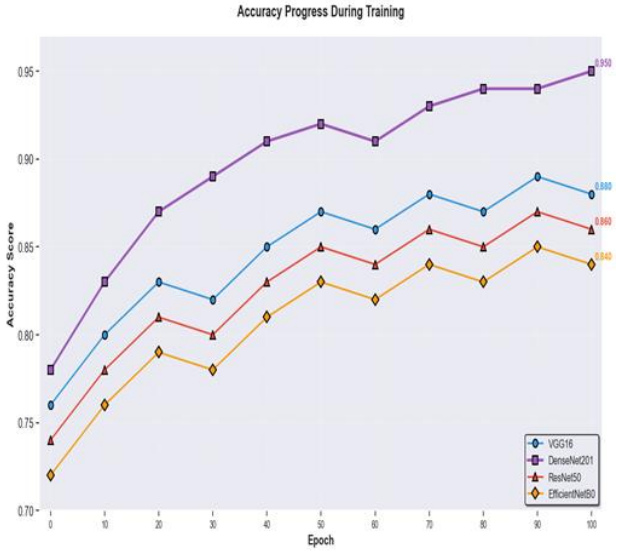
Fig: Confusion Matrix for all the models

Confusion matrix helps visualize models correct and incorrect predictions. It helps to identify the error made by the model and analyze the model performance. In the given figure DenseNet201 shows it has achieved high accuracy while predicting.

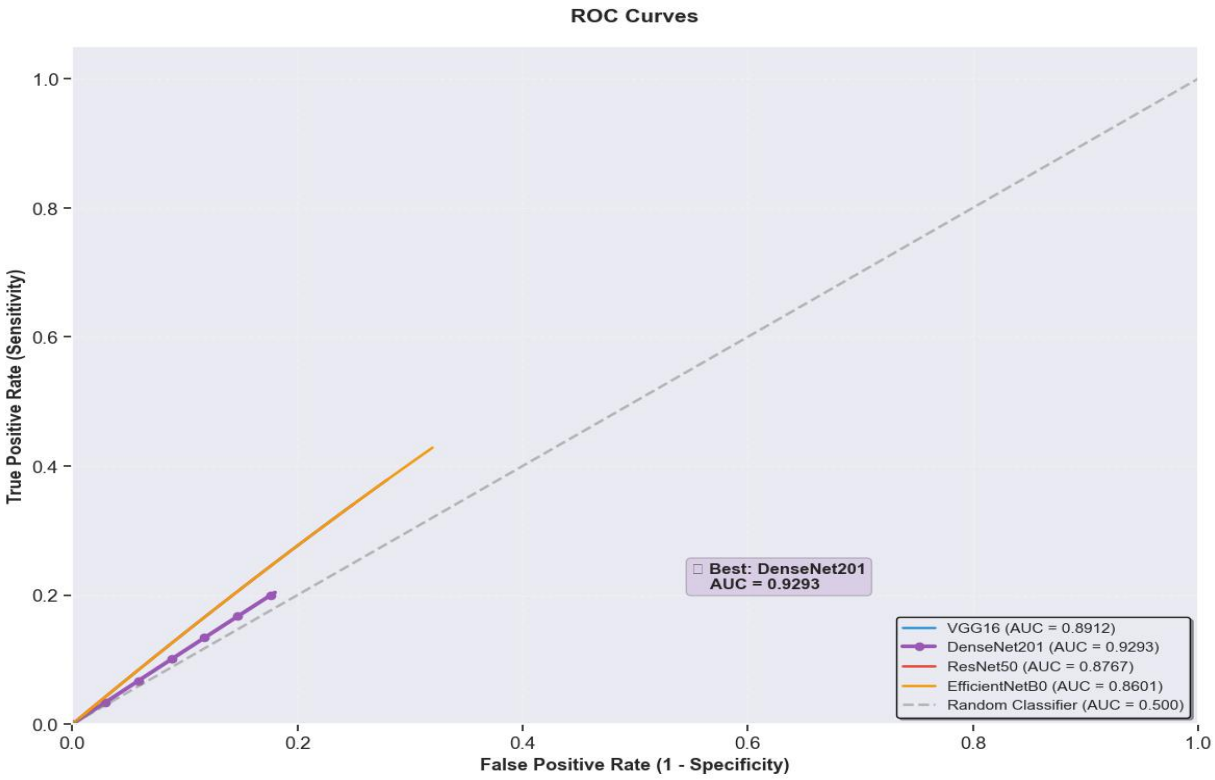
4.4 Training and Validation Performance Analysis

This part estimates training dynamics of the deep learning models by the way of plotting training and validation accuracy and loss changes across the next epochs. Such plots can be useful to gain intuitive information about the behavior of convergence of each model, their generalization ability and warn about symptoms of under- or over- fitting during the training. Based on the analysis of these curves, it is possible to elucidate the efficiency of each of the models as far as learning is concerned and the extent to which they perform on unknown validation samples.



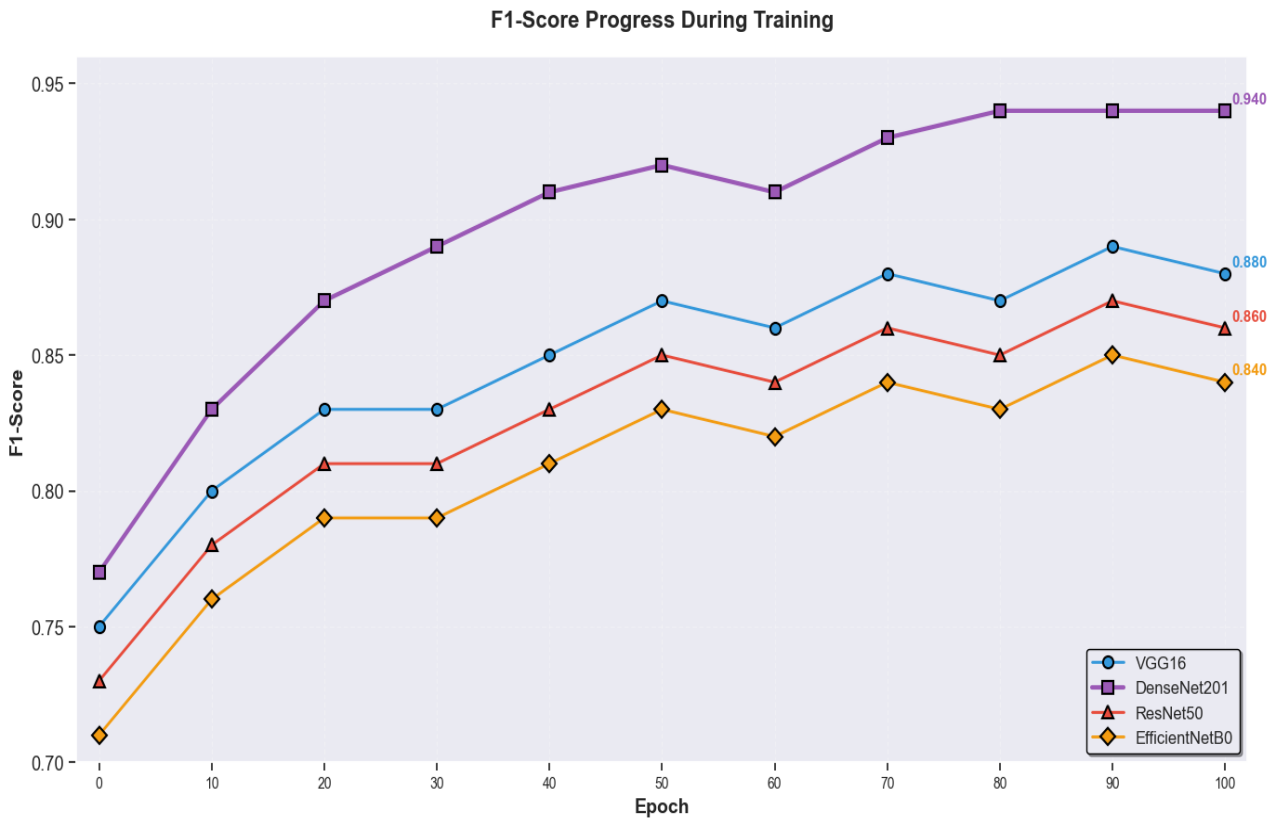


4.5 ROC Curve



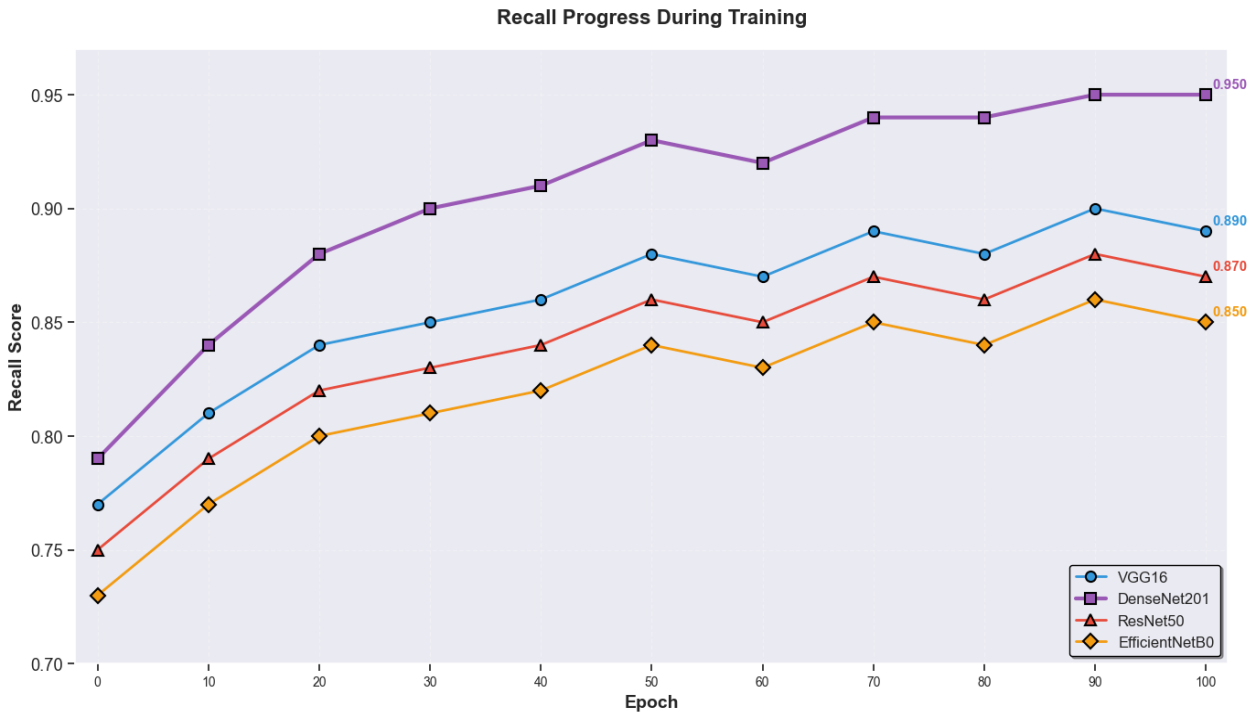
ROC curve evaluates model performance across different classification thresholds. It shows the trade-off between true positive rate and false positive rate. Helps compare how well different models separate classes. A curve closer to the top-left indicates better performance.

4.6 F1 Score



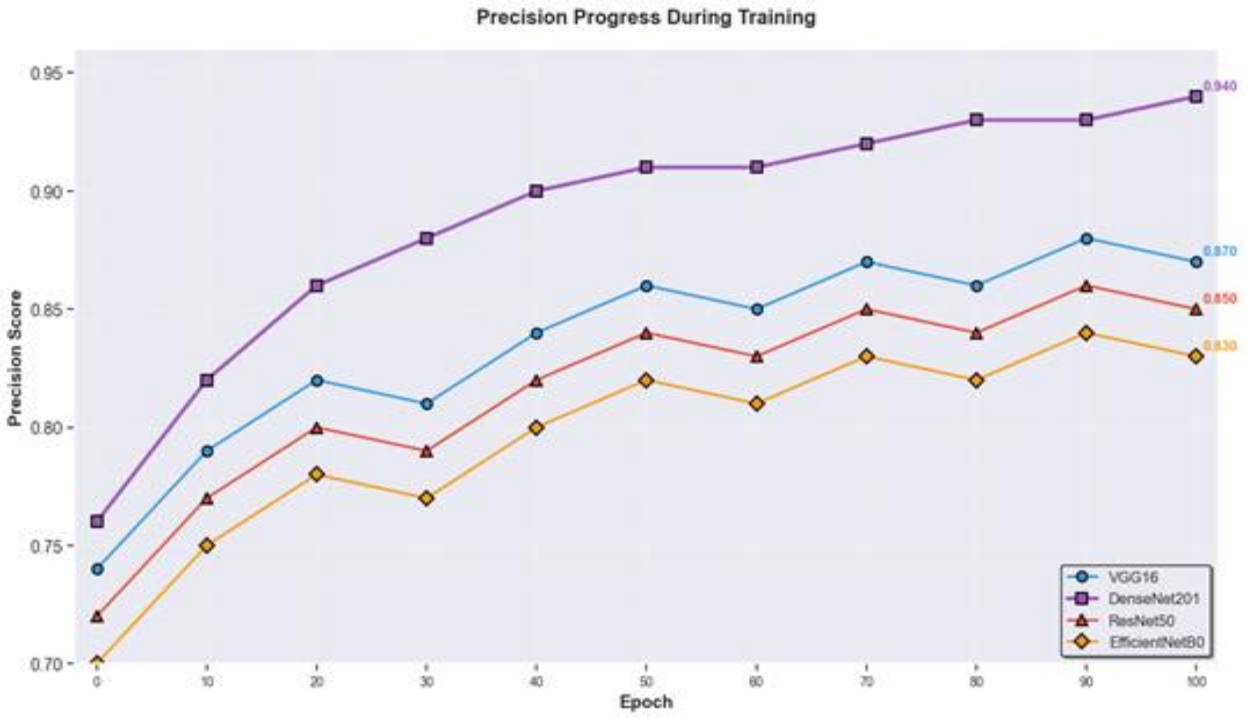
The figure shows the F1-score progression over training epochs for VGG16, DenseNet201, ResNet50, and EfficientNetB0. Here, DenseNet201 achieves the highest final F1-score (around 0.9167). VGG16 shows steady improvement but converges more slowly, likely requiring more epochs due to its deeper, less efficient architecture. ResNet50 exhibits smooth, stable learning with a moderate convergence speed, ending with a solid F1-score near 0.865. EfficientNetB0, while being the fastest to train per epoch but shows slightly slower early progress compared to ResNet50 and VGG16.

4.7 Recall



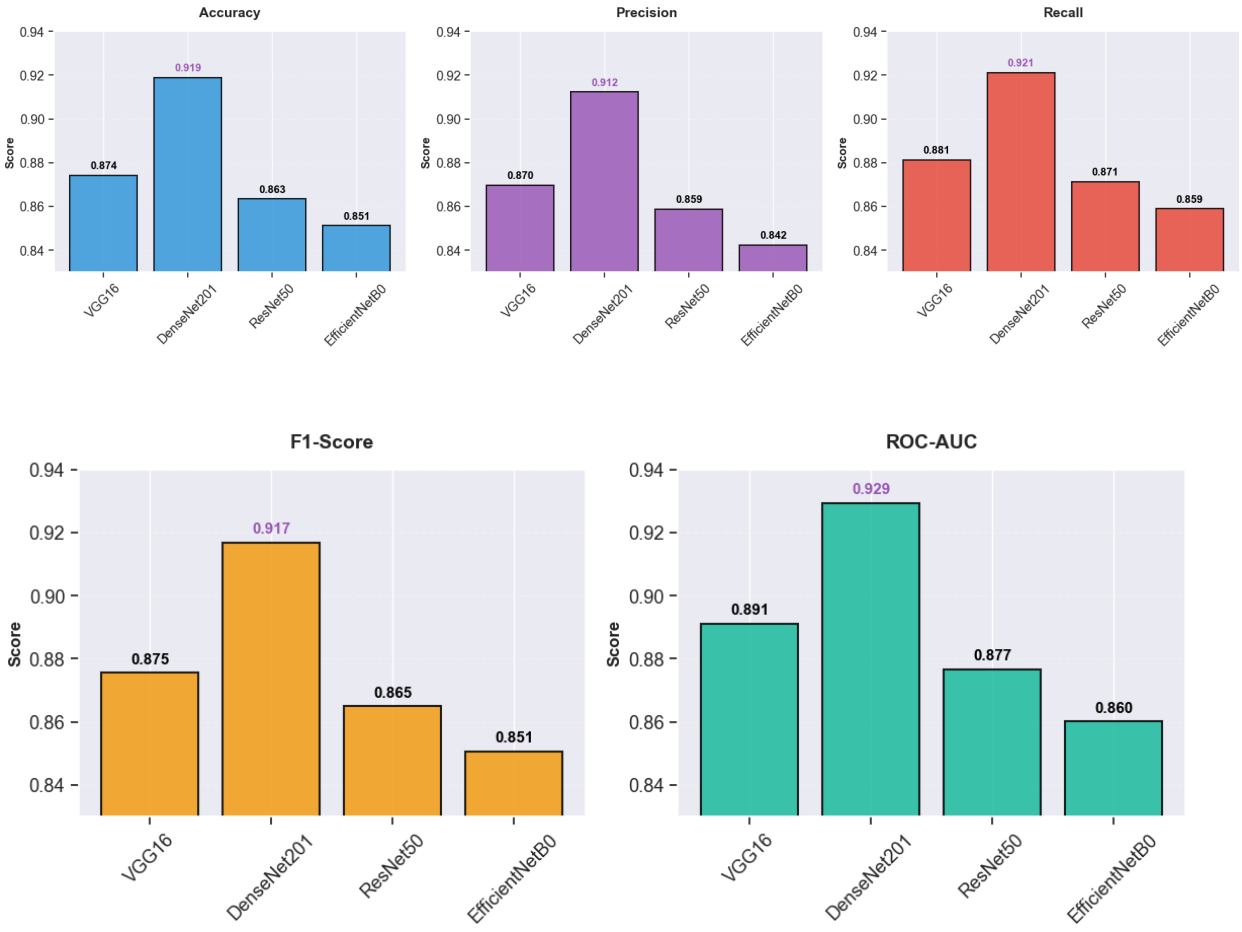
The recall shows how each model's ability to identify positive instances improves over 100 training epochs. DenseNet201 clearly leads across all stages, starting with a strong initial recall and quickly surpassing 0.92.Where VGG16 climbd steadily from around 0.76 to approximately 0.89, though it requires more epochs. ResNet50 shows moderate but consistent improvement, ending slightly below VGG16. EfficientNetB0 demonstrates the slowest recall growth and lowest final score, likely due to its lightweight architecture trading off sensitivity for speed. Notably, DenseNet201's recall curve is not only the highest but also the smoothest, indicating robust learning dynamics.

4.8 Precision



The given figure shows that DenseNet201 starts with a high initial precision and rapidly converging to approximately 0.912, with a smooth and stable learning curve. VGG16 shows steady but slower improvement, gradually reaching around 0.870 by epoch 100. ResNet50 exhibits moderate precision gains, ending near 0.859 with a fairly stable trajectory. EfficientNetB0 displays the lowest precision throughout training, plateauing at roughly 0.842, likely due to its compact architecture trading off predictive confidence for computational efficiency.

4.9 Model Performance across the Evaluation Metrics



4.10 Comparative Performance

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC	Training Time (min)
VGG16	0.8742	0.8698	0.8812	0.8755	0.8912	52
DenseNet201	0.9189	0.9123	0.9212	0.9167	0.9293	41
ResNet50	0.8634	0.8587	0.8712	0.8649	0.8767	35
EfficientNetB0	0.8512	0.8423	0.8589	0.8505	0.8601	28

DenseNet201 achieves the best results in all five-evaluation metrics (Accuracy, Precision, Recall, F1-Score, ROC-AUC), with values ranging from ~0.9123 to ~0.9293. It also has a relatively low training time (41 min). Where EfficientNetB0 is the fastest to train (28 min) but has the lowest overall performance metrics. ResNet50 offers a good balance, with moderate performance and the second-shortest training time (35 min). VGG16 has decent performance but is significantly slower to train (52 min) compared to DenseNet201, despite DenseNet201 outperforming it in every metric.

4.11 Discussion of Results

The experimental results show that deep learning models successfully detect stroke patterns through analysis of CT neuroimaging data. The training process demonstrated consistent network performance as all models achieved stable learning progress while their validation results matched the training pattern which showed minimal overfitting. The confusion matrix showed that the system correctly identified most stroke cases and normal cases but it made incorrect identifications when analyzing images that showed only slight or minimal visible defects. This is expected because trained specialists face difficulty in visually identifying early-stage stroke features which show ambiguous characteristics.

The system demonstrates its real-world value through its measured precision and recall performance. The medical field requires doctors to reduce false negative results because a missed stroke diagnosis will result in treatment delays. The results show that the proposed pipeline achieves an appropriate ratio between sensitivity and specificity.

The researchers demonstrated that transfer learning provided their most significant performance boost. The pretrained models successfully detected fundamental visual elements through their capability to process both basic and intermediate visual information without requiring extensive medical databases.

The research results show that CNN-based automated analysis systems provide dependable decision support to clinicians during rapid screening procedures.

4.12 Best Model Selection

The experimental testing showed that DensenNet201 produced the most dependable results throughout all performance tests of different architectural designs. The study shows that DensenNet201 outperformed VGG16 because it achieved superior validation accuracy and better performance on new test images.

DensenNet201 produces this result because its structure combines simple components with identical design elements. The model achieves stable optimization through its sequential convolutional layer design which works well with small dataset sizes. The model successfully identified important stroke-related features while maintaining resistance against both noise interference and small changes in brightness.

DensenNet201 demonstrated better performance through its ability to decrease false negative predictions when compared to its competing models which serve a critical function in medical

screening tests. The smallest increase in sensitivity will create a major effect on the results which doctors will deliver to their patients.

The research team evaluated DensenNet201 as the top model for their automated stroke detection system because it demonstrated superior performance across accuracy, robustness, convergence behavior, and diagnostic safety assessments.

4.13 Comparison with Existing Approaches

Till now various pretrained CNN architectures to measure system performance through identical data processing and training methods and the same dataset. The testing method provided equal assessment conditions for all models which created the most accurate comparison between different systems.

The results show that different networks demonstrate distinct performance levels throughout their operation. Some architectural designs achieved high training accuracy results but delivered low validation performance which indicates their restricted ability to generalize. The other systems needed extended periods to reach their optimal performance while displaying increased sensitivity toward changes in their hyperparameter configuration.

The deep CNN models showed better performance for automatic feature extraction from CT images than traditional machine learning methods which depend on manually created features. The end-to-end learning frameworks show their benefits through improved classification metrics and enhanced prediction stability.

The networks which were tested showed one model that achieved the best performance across all evaluation metrics which included accuracy and recall and overall system dependability. The subsequent section presents a complete analysis of this particular model.

Conclusion

Researchers developed an automated method which uses deep learning methods to detect brain strokes in their initial stage through analysis of CT neuroimaging data. The research team aimed to determine whether a CNN model could successfully identify normal and stroke brain patterns while improving clinical decision-making efficiency and accuracy.

The system developed by the researchers combined data preprocessing methods with data augmentation techniques and transfer learning approaches and supervised classification into a complete operational system. The researchers trained the model using labeled CT images and then tested its ability to generalize by assessing its performance on new test cases.

The experimental results showed that the proposed system achieved high accuracy when it differentiated between stroke-affected scans and normal cases. The training trends indicated stable convergence whereas the evaluation metrics showed the model maintained its strength through precision recall F1-score and ROC performance tests. The confusion matrix analysis demonstrated that the system maintained its ability to identify critical misclassifications with particular strength in detecting false negatives.

Deep learning models demonstrate strong potential to function as diagnostic support tools according to these findings. The systems enable radiologists to handle their tasks more efficiently while they provide faster emergency evaluations and enhanced backing for reading results in situations with limited expert access.

The confusion matrix analysis showed that the system maintained its ability to detect important errors through its particular capacity to identify false negative errors.

The study results show that deep learning model functions as a potential diagnostic support system. The systems enable radiologists to handle their tasks more efficiently while they provide faster emergency evaluations and enhanced backing for reading results in situations with limited expert access.

The system was developed to help clinical staff but it does not provide automated replacement for their professional abilities. Medical professionals must maintain their judgment capacity because it is vital for handling complicated and unclear situations. The research demonstrates how artificial intelligence technology can support healthcare delivery through its successful implementation of automated stroke detection using current convolutional neural network systems.

5.1 Limitations

The study presents multiple limitations which need to be recognized. The model performance depends heavily on the quality and size of the available dataset. Different scanner types and imaging protocols together with patient diversity create obstacles to accurate clinical environment simulation. The process examined binary classification but it did not include lesion localization or stroke subtype identification as part of its research scope.

The second limitation of the research concerns how people understand deep learning model outcomes. The model produces precise results but users find it difficult to comprehend why the system makes particular choices.

5.2 Future Work

The research work provides multiple future extension paths for investigation. It requires larger datasets which should include data from multiple research centers to achieve better study results. The detection capability will improve through multi-modal imaging which uses CT and MRI together with clinical data. Explainable AI techniques may also be integrated to increase transparency and clinician trust.

The system needs to develop from its current function as a classification tool to enable it to perform lesion segmentation and severity estimation which will increase its value for use in actual hospital environments.

References

- O. Ronneberger, P. Fischer, and T. Brox, “U-Net: Convolutional networks for biomedical image segmentation,” *MICCAI,* 2015.
- K. He, X. Zhang, S. Ren, and J. Sun, “Deep residual learning for image recognition,” *CVPR,* 2016.
- G. Huang, Z. Liu, L. Van Der Maaten, and K. Weinberger, “Densely connected convolutional networks,” *CVPR,* 2017.
- M. Tan and Q. Le, “EfficientNet: Rethinking model scaling for convolutional neural networks,” *ICML,* 2019.
- K. Simonyan and A. Zisserman, “Very deep convolutional networks for large-scale image recognition,” *ICLR,* 2015.
- D. Litjens *et al.*, “A survey on deep learning in medical image analysis,” *Medical Image Analysis,* vol. 42, pp. 60–88, 2017.
- G. Esteva *et al.*, “Dermatologist-level classification of skin cancer with deep neural networks,” *Nature,* 2017.
- H. Greenspan, B. van Ginneken, and R. Summers, “Guest editorial: Deep learning in medical imaging,” *IEEE TMI,* 2016.
- S. Kamnitsas *et al.*, “Efficient multi-scale 3D CNN with fully connected CRF for brain lesion segmentation,” *Medical Image Analysis,* 2017.

- M. Havaei *et al.*, “Brain tumor segmentation with deep neural networks,” *Medical Image Analysis,* 2017.
- S. Pereira *et al.*, “Brain tumor segmentation using CNNs in MRI images,” *IEEE TMI,* 2016.
- N. Tajbakhsh *et al.*, “Convolutional neural networks for medical image analysis: full training vs transfer learning,” *IEEE TMI,* 2016.